



Fig. 2: Underwater robot Artu obeys hand signals (Courtesy: 2.wp.com/www.caddyincadout.com)

g-speak software development kit and runtime libraries, plasma multi-user networking environment, glove based gestural systems and so on can be licensed by customers to fit into their own solutions.

Garnishing education with gestures

Gestures can make education much more effective, claim researchers across the world. Embodied cognition, or learning with the involvement of the whole body, helps understand and assimilate concepts better. Research shows that it reduces cognitive-processing load on the working memory and provides new relations with objects or environments (a psychological concept known as affordances), which are not possible while a person is using only the brain.

For some years now, education researchers have been proposing the use of mixed reality for teaching children difficult concepts. Mixed reality is the merging of real and virtual environments. For example, to learn about the movement of planets, you could have a virtual environment representing the Sun and orbits, while children move around the orbits like planets. Quite

recently, schools have started experimenting with the actual use of embodied cognition and mixed reality, using gesture based interfaces like Kinect, Nintendo Wii and Leap Motion.

When a good tale begins, even the clouds stop and listen. Storytelling is one of the oldest forms of teaching. When an effective storyteller narrates a story, children drop what they are doing and sit around to

listen. If just a one-way story narration can be so effective, imagine how much one can learn through interactive storytelling.

A group of researchers from various universities in China recently got together to develop a gesture based puppetry-assisted storytelling system for young children. The system uses depth-motion-sensing technology to implement a gesture based interface, which enables children to manipulate virtual puppets intuitively using hand movements and other gestures.

What is novel about this system is that it enables multiple players to narrate and to interact with other characters and interactive objects in the virtual story environment. A cloud based network connection is used to achieve this multi-player narration in a virtual environment with low latency.

A user study conducted by the team shows that this system has great potential to stimulate the learning abilities of young children, by promoting creativity, collaboration and intimacy among young children.

Underwater helpers

When you are in deep waters, you would love company. Well, divers,

swimmers and explorers would like some help, too. If you are exploring a sunken ship, for example, a robotic helper that can move heavy chunks on your way or take pictures as instructed could be very useful. European Union project CADDY, or Cognitive Autonomous Diving buddy, is a heavily-funded attempt to build the perfect underwater companion robot to support and monitor human operations and activities during a dive.

CADDY's success depends on the reliability of communication between the diver and the robot. However, underwater environment poses constraints that limit the effectiveness of traditional HMs. This has inspired many research teams to explore gesture based solutions to communicate with CADDY underwater.

Researchers from National Research Council - Institute of Studies on Intelligent Systems for Automation, Genova, Italy, propose to overcome this problem by developing a communication language based on consolidated and standardised diver gestures, commonly employed during professional and recreational dives. This has led to the definition of a CADDY language called CADDIAN and a communication protocol. They have published a research paper that describes the alphabet, syntax and semantics of the language, and are now working on the practical aspects of gesture recognition. Read all about the proposed solution at www.tinyurl.com/j8td6km

Jacobs Robotics Group of Jacobs University in Bremen, Germany, is another group that is researching within CADDY to develop similar solutions for 3D underwater perception to recognise a diver's gestures. Their self-contained stereo camera system with its own processing unit to recognise and interpret gestures was recently tested on Artu, a robot developed by Roman CADDY partner CNR.

Artu is a remotely-operated vehicle that used to be steered via a cable